

Part 7: Logic Operators

1. **Name the basic logic instructions in 8086.** AND, OR, XOR, NOT, TEST.
2. **How does AND work in 8086?** It performs a bitwise AND. The result bit is 1 only if *both* operand bits are 1. It is often used to clear specific bits.
3. **How does OR work?** It performs a bitwise OR. The result bit is 1 if *at least one* of the operand bits is 1. It is often used to set specific bits.
4. **How does XOR work?** It performs a bitwise Exclusive OR. The result bit is 1 if the operand bits are *different*. It is used to toggle bits or clear a register.
5. **How is NOT used?** It inverts every bit in the operand (0 becomes 1, 1 becomes 0). This is also known as the 1's Complement.
6. **Which flags are affected by logic operations?** Sign (SF), Zero (ZF), and Parity (PF) are updated. Carry (CF) and Overflow (OF) are always cleared to 0.
7. **How can you use AND to mask bits?** AND the value with a binary mask where 0s correspond to bits you want to clear (remove) and 1s correspond to bits you want to keep.
8. **How can you toggle bits using XOR?** XOR the value with 1 at the positions you want to flip. $1 \text{ XOR } 1 = 0$ and $0 \text{ XOR } 1 = 1$.
9. **How can you check a specific bit using AND?** AND the value with a mask that has a 1 only at that bit's position. If the Zero Flag (ZF) is set, the bit was 0.
10. **How is SHL used in logic operations?** SHL (Shift Left) shifts bits to the left, filling with 0. It is equivalent to multiplying by 2.
11. **How is SHR used in logic operations?** SHR (Shift Right) shifts bits to the right, filling with 0. It is equivalent to dividing by 2 (unsigned).
12. **How do you perform bitwise inversion?** Use the NOT instruction on the register or memory location.
13. **How do you combine multiple logic operations?** Execute them sequentially. For example, you might AND to clear bits, then OR to set new values.
14. **How can you set or reset a bit in a register?** To **set** (make 1), use OR with a 1 mask. To **reset** (make 0), use AND with a 0 mask at that position.
15. **How is logical comparison done using CMP?** CMP subtracts the source from the destination but discards the result, only updating the flags (like ZF for equality) to be checked by conditional jumps.